

KSP().create()

NOTE: KSP is short hand for Kyrlov subspace methods, which is a method to solve a linear system ($Ax=b$)

Excerpt from code,

```
ksp = PETSc.KSP().create()
ksp.setOperators(K)
ksp.setType("cg")
ksp.getPC().setType("jacobi")
ksp.setFromOptions()
```

The `.KSP.create()` function is a wrapper for the `KSPcreate()` function. The function is,

```
KSPCreate(MPLComm comm, KSP *inksp)
```

Where each of the input and output parameters are,

- `comm` - MPI communicator
- `inksp` - the output parameter

This function produces the settings that the KSP solver will abide by.

Sources from GitHub/Documentations;

- `petsc/petsc/-/blob/release/src/binding/petsc4py/src/petsc4py/PETSc/Mat.pyx`
- `petsc/petsc/-/blob/main/src/mat/impls/aij/seq/aij.c`